

AYUSH UPADHYAY

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SUMMARY

Computer Science (AI & ML) undergraduate, class of 2027, with six software engineering and applied research internships across **Microsoft, Nokia, IIT Ropar, and FlowbitAI Germany**. Ships production systems end to end across the stack: LLM and RAG pipelines, distributed streaming and inference backends, multithreaded C++ systems tooling, and full-stack web applications. Strong foundation in data structures, algorithms, object-oriented design, and scalable system design. **3 IEEE publications, 3 patents, 17 hackathon wins** including 2nd place at Tata InnoVent 2026 across 2,822 teams. Seeking 2027 Software Engineering / AI Engineering internship roles.

EDUCATION

B.Tech, Computer Science and Engineering (AI & Machine Learning)

2023 – 2027

Vellore Institute of Technology, Chennai, India

- **Coursework:** Data Structures and Algorithms, Operating Systems, Computer Networks, Database Management Systems, Object-Oriented Programming, Machine Learning, Computer Architecture, Software Engineering
- **Leadership:** Founder, [HackFinity](#) (1,550+ participants, 65+ countries) • Chairperson, [Institution of Engineers India \(IEI\)](#) VIT Chennai • Technical Lead, ACM Student Chapter

EXPERIENCE

Applied Scientist Intern | Microsoft, Hyderabad, India

May 2026 – Jul 2026

- Built and shipped a cohort-based recommendation engine for Microsoft 365 Copilot upskilling inside Viva Learning, clustering users on behavioral and skill signals to serve adaptive course recommendations from a user's first day of activity.
- Designed the cohort-vector schema and a five-generation synthetic data pipeline with automated quality validation, registering datasets and training runs on Azure ML.
- Benchmarked nine clustering approaches in a segmentation prototype, then integrated the selected model into AILA, the product surface consuming the engine.

Full Stack AI Developer | Tenchi Consultation, Dubai, UAE

Jan 2026 – Mar 2026

- Architected SmartOrder Engine, an AI order-management layer over SAP S/4HANA serving FMCG distributors across the UAE, Qatar, and Saudi Arabia.
- Delivered automated invoice OCR, real-time inventory tracking, and Gemini 2.0 Flash demand forecasting behind a REST API consumed by client ERP workflows.

AI Full Stack Developer | FlowbitAI, Germany

Jan 2026 – Mar 2026

- Redesigned the document OCR pipeline around a CNN-Transformer hybrid, raising extraction accuracy 35% while cutting inference cost 40%.
- Rebuilt the backend as an async FastAPI service with Redis queuing and AWS Lambda workers, holding sub-2-second latency at 50,000+ pages per month.

AI Developer Intern | Nokia

May 2025 – Aug 2025

- Deployed XGBoost and LSTM predictive-maintenance models across 12 5G/6G network clusters, reducing unplanned downtime 30%.
- Built event-driven Apache Kafka streaming pipelines ingesting 1M+ telemetry records daily into the feature store powering those models.

AI/ML Research Intern | Annam AI, IIT Ropar

Apr 2025 – Nov 2025

- Trained computer vision models in TensorFlow and PyTorch to 92% accuracy on limited-sample agricultural datasets.
- Accelerated training convergence 40% through transfer learning and augmentation strategies on pretrained backbones.

AI Software Engineer Intern | Real IT Solutions

May 2025 – Jul 2025

- Shipped full-stack AI applications in React, Node.js, and FastAPI sustaining 99.9% uptime.
- Automated CI/CD with GitHub Actions and Git-based code review workflows, cutting deployment cycle time 85%.

PROJECTS

[NeuroDrive — Agentic Self-Healing EV Ecosystem](#) — Python · Agentic AI · AWS · IoT

- 2nd place at Tata InnoVent 2026 among 2,822 teams and 10,247 innovators. Autonomous EV diagnostics with an agentic self-healing framework that cut simulated fleet downtime 65%, plus real-time fleet monitoring on AWS.

[Packet Analyzer — Deep Packet Inspection Engine](#) — C++ · Python · Multithreading · TLS/SNI · BPF

- Multithreaded DPI engine with SNI/TLS extraction, BPF-filtered capture, per-flow state tracking, and a load-balanced worker pipeline for live traffic classification.

[CrispRoots — AI AgroTech Platform](#) — Machine Learning · Computer Vision · IoT · React

- Won Rs 2L at the IIT Ropar Annam AI Hackathon plus Rs 1L seed funding from Balkalman Funds. Precision agriculture platform with crop disease detection, soil analysis, and yield prediction.

[GyanGrow — Gamified Financial Literacy Platform](#) — *React · Node.js · TensorFlow · MongoDB*

- 50,000+ users and 100+ quests with AI-personalized learning roadmaps, tracking Rs 2Cr+ in virtual savings at a 95% quest completion rate.

[Echo-Med-AI — Mobile Medical Diagnostics](#) — *TensorFlow · OpenCV · CNN · TFLite*

- CNN diagnostic tool reaching 94% accuracy across six disease categories, optimized for mobile inference via TFLite quantization with a 75% reduction in model size.

[AuraTwin — AI Infrastructure Safety](#) — *AWS Bedrock · Gaussian Splatting · 3D Reconstruction*

- Entry in the AWS 10,000 Aldeas Competition. Phone-based structural inspection using Gaussian Splatting for 3D reconstruction and Amazon Bedrock for real-time risk analysis.

TECHNICAL SKILLS

Languages: Python, C++, C, Java, JavaScript, TypeScript, Go, Rust, Kotlin, SQL, Bash, HTML/CSS

CS Fundamentals: Data Structures, Algorithms, Dynamic Programming, Graph Algorithms, Recursion and Backtracking, Time and Space Complexity Analysis, Object-Oriented Design, Design Patterns, Operating Systems, Computer Networks, DBMS, Compiler Basics

System Design: High-Level and Low-Level Design, Distributed Systems, Scalability, Sharding, Replication, Consistent Hashing, CAP Theorem, Load Balancing, Caching Strategies, API Gateways, Message Queues, Event-Driven Architecture, Rate Limiting, Fault Tolerance, Concurrency and Multithreading, Low-Latency Optimization

AI / ML: PyTorch, TensorFlow, Keras, scikit-learn, XGBoost, OpenCV, Transformers (BERT, GPT), Hugging Face, CNNs, LSTMs, Feature Engineering, Hyperparameter Tuning, Transfer Learning, Model Evaluation, Quantization, Model Deployment, MLOps

LLM and GenAI: LangChain, LangGraph, LlamaIndex, RAG, Vector Databases, Embeddings, Semantic Search, Fine-Tuning, LoRA/QLoRA, Prompt Engineering, LLM Orchestration, AI Agents, Agentic AI, Voice AI (ASR/TTS), NLP, Computer Vision

Backend and Web: React, Next.js, Node.js, Express, FastAPI, Django, Flask, REST API Design, GraphQL, Microservices, WebSockets, JWT and OAuth Authentication, Rate Limiting, Async Programming, Server-Side Rendering

Databases: PostgreSQL, MySQL, MongoDB, Redis, Elasticsearch, Pinecone, Query Optimization, Indexing, Schema Design, Data Modeling

Cloud: AWS (EC2, S3, Lambda, SageMaker, Bedrock, IAM, CloudWatch, API Gateway), Azure (Azure ML, Blob Storage), Serverless Architecture, Cloud Cost Optimization

DevOps and Infrastructure: Docker, Kubernetes, CI/CD Pipelines, GitHub Actions, Jenkins, Nginx, Terraform, Infrastructure as Code, Container Orchestration, Apache Kafka, Monitoring and Observability, Prometheus, Grafana, Log Aggregation, Blue-Green Deployment, Linux, Shell Scripting, Git and GitHub, Version Control

Testing and Quality: Unit Testing, Integration Testing, pytest, Jest, Debugging, Profiling, Code Review, Test-Driven Development

Data and Tools: Microsoft Excel (advanced formulas, pivot tables, VLOOKUP, data modeling, dashboards), Pandas, NumPy, Matplotlib, Jupyter, Power BI, Postman, SAP S/4HANA, Figma, Jira, Confluence

Practices: Agile and Scrum, Sprint Planning, Technical Documentation, Cross-Functional Collaboration, Stakeholder Communication, Mentoring, Problem Solving, Ownership

PUBLICATIONS AND PATENTS

- **Patent (Granted, India) — No. 202541101828 A** — "Regenerative Power System for Vehicles," an AI-orchestrated quantum-tunnelling braking system.
- **Patent (Filing in progress)** — Advanced Selective Active Noise Cancellation System: adaptive, selective suppression of targeted noise bands.
- **Patent (Filing in progress)** — BB-KDF, a Behavioral Biometric Key Derivation Function deriving ephemeral AES-256 session keys from live keystroke, pointer, touch, and gaze signals via LSTM, Reed-Solomon, and HKDF.
- **IEEE CICT 2025** — "Efficient Pruning in Subset Sum Backtracking" (Paper ID CICT2025-540).
- **IEEE** — "AI-Driven Resource Optimization for 5G/6G Heterogeneous Networks": spectrum allocation and predictive load balancing.
- **IEEE** — "Solid-State Battery Degradation Prediction via Hybrid ML": LSTM-attention for cycle-life estimation.

ACHIEVEMENTS AND LEADERSHIP

- **17x Hackathon Winner** — [Tata InnoVent 2026](#) (2nd place, 2,822 teams), [Samsung Solve for Tomorrow](#), [Annam AI, IIT Ropar](#), [HackHazards](#), [AWS 10K Aldeas](#), IntelliTrace (Indian Bank), Garage Day, and 10 more.
- **Innovation Ambassador**, Ministry of Education, Government of India.
- Organizer, [Nexora Hacks 2026](#) — \$7,500 prize pool, 695 participants, 32 judges from Apple, Meta, Netflix, AWS, and Walmart.
- **Certifications:** Generative AI (IBM) • AI Fundamentals (Microsoft) • Full Stack Developer (Coursera) • AI Developer (AMD) • AWS Cloud Practitioner • GitHub Professional • LangChain and LangGraph • Backend Development.